

Hospital Bed Allocation and Patient Flow Simulation

A Discrete-Event Simulation Study of Capacity and Queue-Policy Interventions

Department of Mathematics and Computer Science
Iran University of Science and Technology

Course: Computer Simulation
Instructor: Dr. Rahman Farnoosh

Students:
Arshia Samimi
Bahar Jafarnejad

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Abstract

We build a discrete-event simulation of a 10-bed inpatient unit with a finite queue, covering patient arrival, triage, bed allocation, treatment, and discharge, over a 30-day (720-hour) horizon. Using 30 independent replications per scenario, we compare the base case against three interventions — additional bed capacity, a priority-based queue discipline, and their combination — and stress-test all of them under a higher-variance (exponential) arrival process. All comparisons are backed by 95% confidence intervals across replications and a formal practical-significance test. The combined intervention (12 beds + priority queue) is the strongest performer on rejection rate and waiting time under both arrival regimes, though its benefit over capacity alone narrows considerably once arrivals become bursty.

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1 Executive Summary

Under the base configuration (10 beds, queue capacity 5, FIFO, uniform(1,3)h arrivals), the hospital rejects roughly 15% of arriving patients and keeps beds occupied about 99% of the time — indicating the base case is already capacity-constrained rather than comfortably provisioned. Three interventions were tested against this baseline:

- **Capacity expansion** to 12 beds (selected via a diminishing-returns analysis over 10–24 beds) cuts rejection to approximately 1.7%.
- **A priority-based queue discipline** (highest priority score served first, instead of FIFO) leaves the overall rejection rate statistically unchanged but reallocates who gets rejected — shifting the burden away from high-priority patients and onto low-priority ones.
- **Combining both** interventions produces the best results on nearly every tracked metric under normal (uniform-arrival) conditions, but the added value of the priority queue on top of extra capacity shrinks sharply once arrivals become bursty (exponential inter-arrival times), where capacity alone accounts for almost all of the improvement.

Our recommendation, detailed in Section 8, is to expand to 12 beds as the primary intervention, with the priority queue retained as a low-cost clinical-fairness safeguard rather than a throughput lever.

2 Model Assumptions

2.1 Simulation Horizon and Event Types

The simulation runs for 30 days (720 hours), driven by six event types: `ARRIVAL`, `TRIAGE_DONE`, `QUEUE_TIMEOUT`, `DISCHARGE`, and hourly `MONITOR` events, processed in a single priority-queue-based future event list (FEL) ordered by (time, insertion order).

Horizon convention (censoring). New patient arrivals stop being scheduled once the next arrival time would fall at or beyond 720 hours, but the event loop continues draining the FEL until every already-arrived patient is fully resolved (admitted and discharged, or rejected). No patient is censored: every admitted patient in our data has a recorded discharge time. This choice was verified programmatically for every replication (Section 3.2).

2.2 Patient Attributes

Age groups and **diseases** are drawn per Appendix A of the assignment (Table 1 and Table 2); subtypes within a disease are drawn uniformly. Blood pressure, pulse, and temperature are drawn from clipped normal distributions ($\mathcal{N}(120, 15)$ mmHg, $\mathcal{N}(80, 15)$ bpm, $\mathcal{N}(37.0, 0.6)$ °C respectively, clipped to physiologically plausible ranges) as a reasonable proxy for a patient population that is mostly normal with occasional abnormal readings.

Table 1: Age group distribution and priority weight (Appendix A).

Age group	Probability	Age priority	Factors
0–18	0.15	3	pediatric
19–35	0.25	2	young adult
36–55	0.35	1	adult
56–75	0.15	4	elderly, comorbidities
76+	0.10	5	geriatric, high risk

Table 2: Disease table: base severity and treatment-duration range (Appendix A).

Disease	Base severity	Treatment range
Broken Arm	2	12–24 h
Concussion	4	18–24 h
Simple Fracture	3	12–18 h
Appendicitis	5	12–24 h
Pneumonia	4	18–24 h

2.3 Severity, Priority, and Waiting-Time Formulas

Severity score. Subtype modifiers are $-1/0/+1$ for the three subtypes of each disease (mild to severe), and abnormal vitals (BP outside 90–140 mmHg, pulse outside 60–100 bpm, or temperature outside 36.1–37.8°C) each add one point:

$$\text{severity} = \text{base_severity} + \text{subtype_modifier} + \text{n_abnormal_vitals} + 0.5(\text{age_priority} - 1).$$

Priority score and level. Severity is combined with a separate age-priority term so that both clinical severity and age-related vulnerability matter (age is consequently weighted twice — once inside severity via the age factor, once again here — a deliberate design choice to give age extra weight in triage, noted here for transparency):

$$\text{priority} = 0.7 \text{severity} + 0.3(2 \times \text{age_priority}), \quad \text{level} = \begin{cases} \text{high,} & \text{priority} \geq 8 \\ \text{medium,} & 4 \leq \text{priority} < 8 \\ \text{low,} & \text{priority} < 4. \end{cases}$$

Safe waiting time. Higher-priority patients are given a shorter safe waiting time in the queue, as required:

$$\text{safe_wait} = \max(1.5, 12 - 0.5 \times \text{priority}) \quad [\text{hours}].$$

A queued patient who exceeds this limit is rejected with reason `timeout`.

Treatment duration. Drawn from a triangular distribution over the disease’s range (mode at the midpoint) and scaled by severity, so sicker patients occupy a bed longer:

$$\text{duration} = \text{Triangular}(\text{low}, \text{mode}, \text{high}) \times (0.8 + 0.1 \times \text{severity}).$$

2.4 Rejection Reasons

Two rejection paths are implemented, matching the two rejection branches in the assignment’s flowchart: `queue_full` (bed unavailable at triage *and* the 5-slot queue is already full) and `timeout` (a queued patient exceeds their safe waiting time before a bed frees up).

3 Implementation

3.1 Discrete-Event Architecture

The simulation is a classical next-event time-advance engine: a min-heap future event list holds `(time, sequence, event_type, patient_id)` tuples; the main loop pops the earliest event, dispatches to the matching handler, and lets that handler schedule any resulting future events. Patient flow follows the required diagram exactly: `ARRIVAL` $\rightarrow$ `TRIAGE` $\rightarrow$ (bed available? admit : queue-if-space-else-reject) $\rightarrow$ (queued patient either gets a bed before timeout, or is rejected) $\rightarrow$ `TREATMENT` $\rightarrow$ `DISCHARGE`, at which point the next eligible queued patient (FIFO or priority order, depending on scenario) is immediately admitted.

3.2 Verification

Every replication of every scenario is checked programmatically against the minimum verification checklist (Appendix B of the assignment) before its results are accepted:

- No bed is occupied by more than one patient at the same time.
- Occupied beds never exceed bed capacity; queue length never exceeds queue capacity.
- Every admitted patient has a bed number, admission time, and discharge time.
- Every rejected patient has a rejection time and rejection reason.
- Replications use distinct random seeds (seeded per scenario and per replication index).

Across all scenarios and all 30-replication batches run for this report, **zero verification issues were found**. A representative single base-case run simulated 371 patients (309 admitted, 62 rejected — 43 by timeout, 19 by queue-full) with 0 issues.

4 Statistical Methodology

- Each scenario is run for **30 independent replications**, each with a distinct random seed (seeded per scenario *and* per replication index, so no two runs anywhere in this study share a random stream).
- For each replication, the 9 required performance metrics (Section 6.1) are computed from that run’s patient-level and hourly monitoring tables.
- Reported means and 95% confidence intervals are computed **across the 30 replication-level metric values**, using a *t*-distribution critical value (not computed by pooling individual patients from a single run).
- **Practical significance** between any two scenarios is declared only when *both* (a) the 95% CIs for that metric do not overlap, and (b) the gap between means exceeds a domain-meaningful threshold (5 percentage points for rate/proportion metrics, 1 hour for time metrics). This avoids treating a statistically detectable but operationally trivial difference as a real finding.

5 Experiment Design

All experiments start from the unmodified base case: 10 beds, queue capacity 5, FIFO discipline, uniform(1,3)h inter-arrival times.

5.1 Experiment 1 — Capacity Expansion

Change. Bed count swept over {10, 11, 12, 14, 16, 18, 20, 24}, all else held at base-case values.

Justification. Capacity is the most direct lever against a system that is already near saturation (99% bed utilization at baseline); sweeping a wide range lets us locate the point of diminishing returns rather than picking a number arbitrarily. **Expected effect.** Rejection rate should fall monotonically with more beds, with returns diminishing as the system moves away from saturation.

Selection method. Walking up the swept bed counts, each successive increment is kept only if it is practically significant (non-overlapping CIs and >5 percentage-point drop in rejection rate) versus the previous increment; the walk stops at the first non-significant step. **Result.**

Step	Gap in mean rejection rate	Practically meaningful?
10 → 11 beds	7.44 pp	Yes
11 → 12 beds	5.16 pp	Yes
12 → 14 beds	1.75 pp	No — stop

12 beds was selected and carried forward into Experiments 3 and 4.

5.2 Experiment 2 — Queue Policy

Change. Replace FIFO queue discipline with a priority rule: the queued patient with the highest priority score is admitted first (ties broken by earlier queue-entry time), still at 10 beds. **Justi-**

fication. FIFO ignores clinical urgency; a priority rule directly targets fairness for high-severity patients without spending any additional capacity. **Expected effect.** Overall rejection rate should change little (the same finite number of beds and queue slots are still being consumed), but the *composition* of who is rejected should shift toward lower-priority patients. **Result.** Confirmed — see Section 7.1: overall rejection rate is statistically unchanged, but low-priority rejection rises while high- and medium-priority rejection fall.

5.3 Experiment 3 — Combined (Capacity × Queue Policy)

Change. 12 beds (from Experiment 1) *and* the priority queue (from Experiment 2) applied together. **Justification.** Tests whether the two interventions are complementary (capacity relieves the overall bottleneck; the queue policy governs who benefits from it) or redundant. **Expected effect.** Best performance on essentially every metric, since it inherits both benefits. **Result.** Confirmed under uniform arrivals (Section 7.1); notably weaker incremental benefit over capacity-alone under exponential arrivals (Section 7.2).

5.4 Experiment 4 — Demand Stress Test (Exponential Arrivals)

Change. Inter-arrival times switched from uniform(1,3)h (mean 2h, low variance) to exponential(mean 2h) (same mean, much higher variance/burstiness), applied to all four scenarios above (base, 12 beds, priority queue, combined). **Justification.** Real hospital demand is bursty, not evenly spaced; testing the same mean arrival rate under higher variance checks whether conclusions

from the uniform-arrival experiments are robust to more realistic demand patterns. **Expected effect.** All scenarios should perform worse (higher rejection, longer waits) than their uniform-arrival counterparts, since queueing systems are generally variance-sensitive even at a fixed mean load. **Result.** Confirmed; see Section 7.2 for the full comparison.

6 Results

6.1 Required Performance Metrics

Metric	Definition
Admission rate	admitted / total patients
Rejection rate	rejected / total patients
Average waiting time	mean queue-entry-to-admission time (admitted patients)
Average bed utilization	mean occupied beds / total beds, over monitoring times
Peak queue length	maximum observed queue length
Percent time queue full	fraction of monitoring times at queue capacity
Percent time under pressure	fraction of times (beds used + queue length) > bed capacity
Average length of stay	mean discharge time – arrival time (admitted patients)
Rejection rate by priority	low / medium / high priority rejection rates

6.2 Base Case — Full Metric Summary (Mean $\pm$ 95% CI, 30 replications)

Table 3: Base case: 10 beds, queue capacity 5, FIFO, uniform(1,3)h arrivals.

Metric	Mean	95% CI Low	95% CI High	n
admission rate	85.22%	84.65%	85.80%	30
rejection rate	14.78%	14.20%	15.35%	30
avg waiting time	5.48 h	5.39 h	5.57 h	30
avg bed utilization	98.55%	98.49%	98.62%	30
peak queue length	5.00	5.00	5.00	30
pct time queue full	9.79%	9.22%	10.35%	30
pct time under pressure	92.77%	92.10%	93.44%	30
avg length of stay	29.31 h	29.14 h	29.48 h	30
rejection rate low	10.05%	9.18%	10.92%	30
rejection rate medium	16.74%	15.92%	17.56%	30
rejection rate high	26.43%	19.25%	33.62%	30

6.3 Required Visualizations

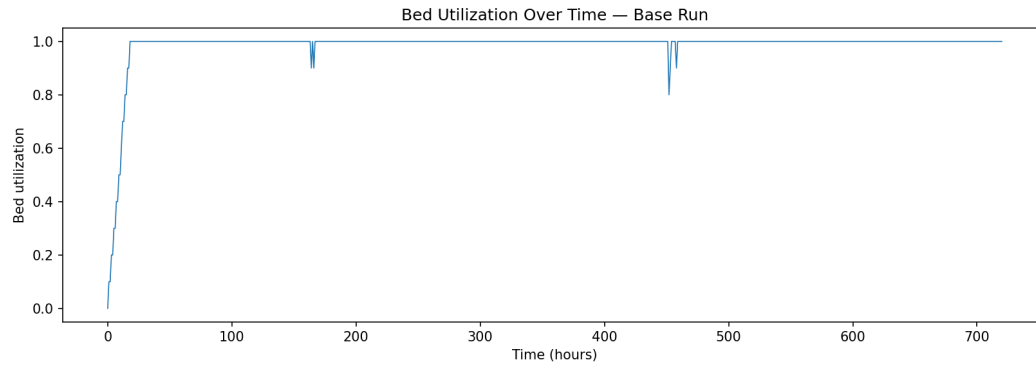


Figure 1: Bed utilization over time, representative base-case run.



Figure 2: Queue length over time, representative base-case run.

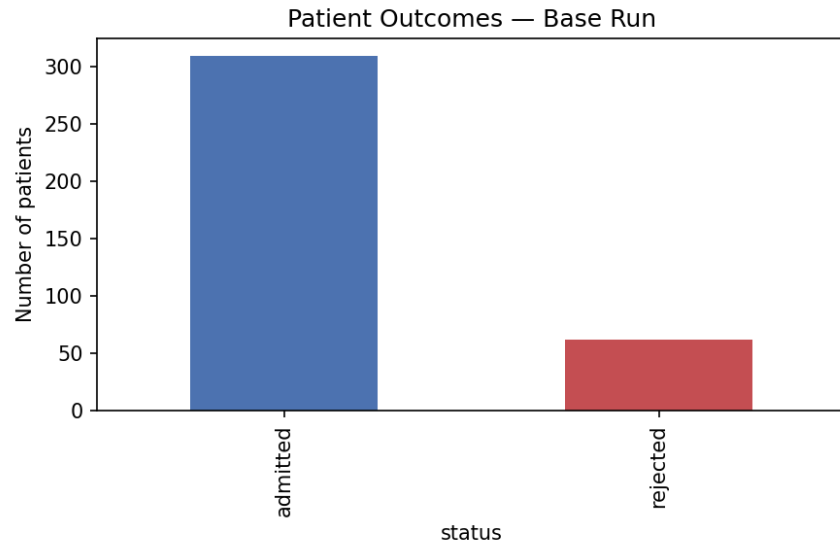


Figure 3: Patient outcomes: admitted vs. rejected, representative base-case run. (No patients are censored under the drain-to-completion horizon convention used here.)

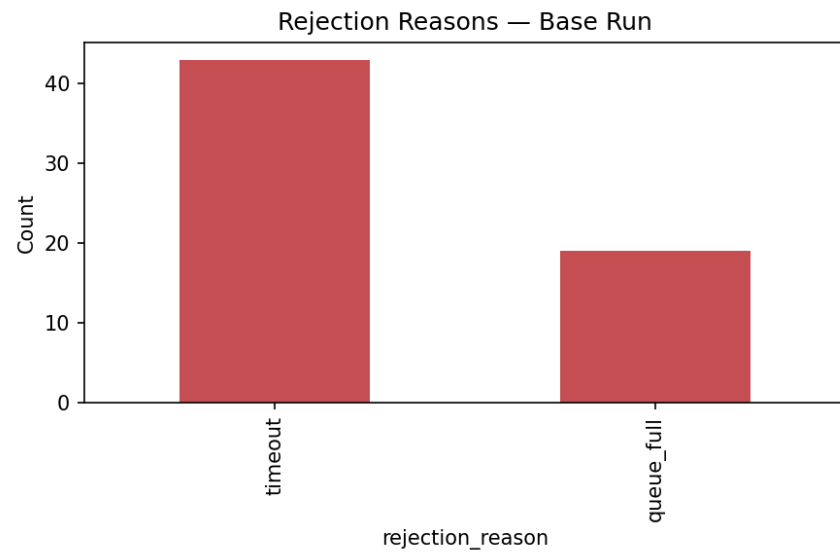


Figure 4: Breakdown of rejection reasons, representative base-case run.

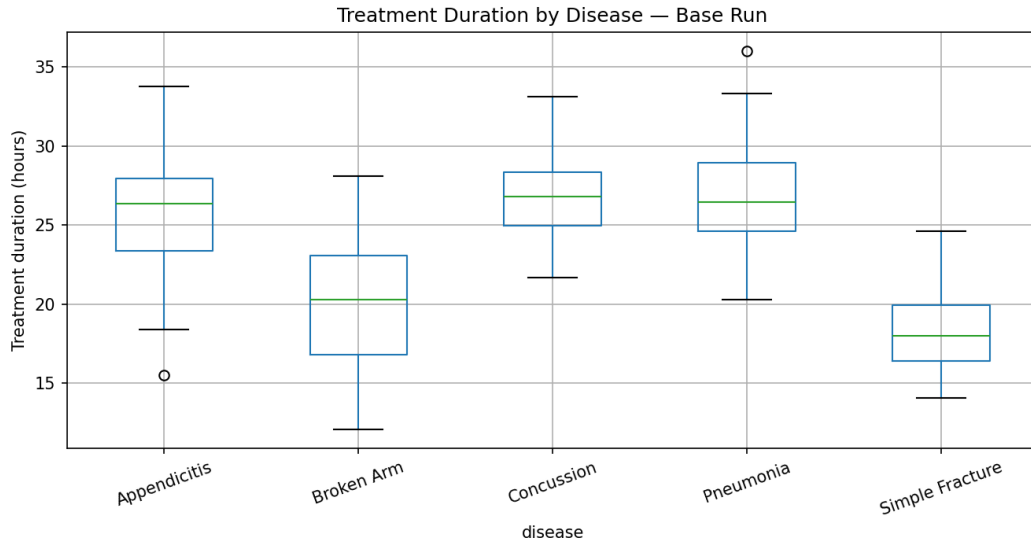


Figure 5: Treatment duration by disease.

7 Scenario Comparison and Confidence Intervals

This section compares the four core scenarios — **Base** (10 beds, FIFO), **12 beds** (capacity alone), **Priority** (queue policy alone), and **Combined** (12 beds + priority queue) — first under the base uniform-arrival process, then, separately, under the exponential-arrival stress test. The two arrival regimes are kept in fully separate tables/figures throughout, per design.

7.1 Uniform Arrivals

Table 4: All four scenarios, all 11 tracked metrics, mean and 95% CI across 30 replications each (uniform arrivals). Rows are grouped by metric, with scenarios in the order base / 12 beds / priority / combined.

Scenario–Metric	Mean	Low	High	n
admission rate base	85.22%	84.65%	85.80%	30.0
admission rate 12beds	98.25%	97.77%	98.74%	30.0
admission rate priority	82.47%	81.71%	83.23%	30.0
admission rate combined	97.35%	96.96%	97.74%	30.0
rejection rate base	14.78%	14.20%	15.35%	30.0
rejection rate 12beds	1.75%	1.26%	2.23%	30.0
rejection rate priority	17.53%	16.77%	18.29%	30.0
rejection rate combined	2.65%	2.26%	3.04%	30.0
avg waiting time base	5.48 h	5.39 h	5.57 h	30.0
avg waiting time 12beds	2.53 h	2.28 h	2.77 h	30.0
avg waiting time priority	2.72 h	2.67 h	2.77 h	30.0
avg waiting time combined	1.21 h	1.12 h	1.31 h	30.0
avg bed utilization base	98.55%	98.49%	98.62%	30.0

Scenario–Metric	Mean	Low	High	<i>n</i>
avg bed utilization 12beds	95.92%	95.55%	96.30%	30.0
avg bed utilization priority	98.20%	98.11%	98.29%	30.0
avg bed utilization combined	94.55%	94.13%	94.98%	30.0
peak queue length base	5.00	5.00	5.00	30.0
peak queue length 12beds	4.97	4.90	5.03	30.0
peak queue length priority	5.00	5.00	5.00	30.0
peak queue length combined	4.57	4.38	4.75	30.0
pct time queue full base	9.79%	9.22%	10.35%	30.0
pct time queue full 12beds	1.58%	1.09%	2.08%	30.0
pct time queue full priority	2.72%	2.39%	3.05%	30.0
pct time queue full combined	0.26%	0.15%	0.38%	30.0
pct time under pressure base	92.77%	92.10%	93.44%	30.0
pct time under pressure 12beds	60.14%	56.33%	63.96%	30.0
pct time under pressure priority	83.48%	82.54%	84.42%	30.0
pct time under pressure combined	42.74%	39.99%	45.49%	30.0
avg length of stay base	29.31 h	29.14 h	29.48 h	30.0
avg length of stay 12beds	26.59 h	26.28 h	26.89 h	30.0
avg length of stay priority	27.32 h	27.17 h	27.47 h	30.0
avg length of stay combined	25.30 h	25.13 h	25.47 h	30.0
rejection rate low base	10.05%	9.18%	10.92%	30.0
rejection rate low 12beds	1.20%	0.63%	1.77%	30.0
rejection rate low priority	36.72%	35.09%	38.35%	30.0
rejection rate low combined	6.05%	5.03%	7.08%	30.0
rejection rate medium base	16.74%	15.92%	17.56%	30.0
rejection rate medium 12beds	1.83%	1.29%	2.37%	30.0
rejection rate medium priority	8.84%	8.19%	9.49%	30.0
rejection rate medium combined	1.09%	0.80%	1.37%	30.0
rejection rate high base	26.43%	19.25%	33.62%	30.0
rejection rate high 12beds	6.65%	3.35%	9.94%	30.0
rejection rate high priority	3.49%	1.15%	5.83%	30.0
rejection rate high combined	0.00%	0.00%	0.00%	30.0

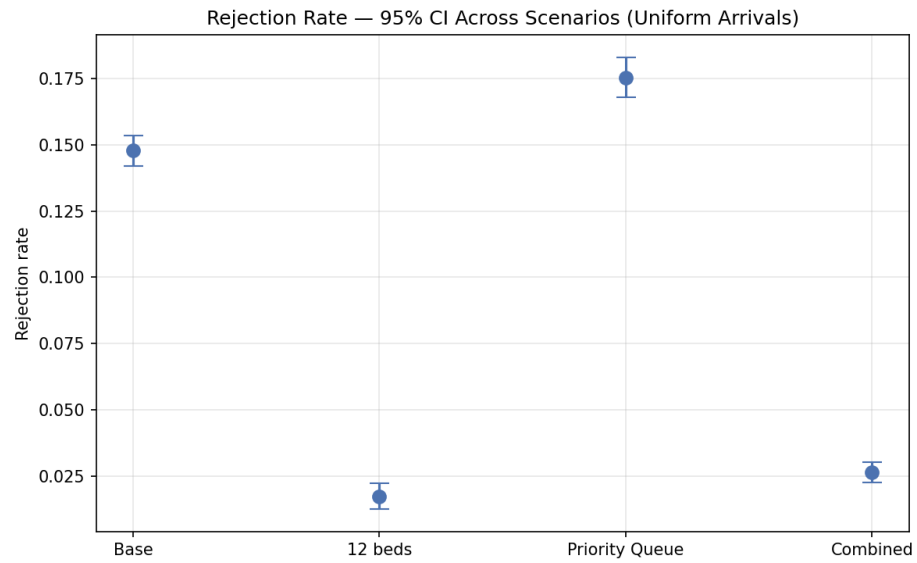


Figure 6: Rejection rate, mean and 95% CI, across all four scenarios (uniform arrivals).

Table 5: Practical significance matrix (uniform arrivals). Rows are metrics, columns are scenario comparisons. “–” means the difference is *not* practically meaningful (95% CIs overlap, or the gap is below the domain threshold); otherwise the cell names which scenario is better and by how much. Higher is better for admission rate and bed utilization; lower is better for every other metric.

Metric	Combined vs. Base	Combined vs. Capacity-alone	Combined vs. Queue-Policy-alone	Queue Policy vs. Base
admission rate	combined better by 12.1pp	-	combined better by 14.9pp	-
rejection rate	combined better by 12.1pp	-	combined better by 14.9pp	-
avg waiting time	combined better by 4.27h	combined better by 1.31h	combined better by 1.50h	priority better by 2.76h
avg bed utilization	-	-	-	-
peak queue length	combined better by 0.43	combined better by 0.40	combined better by 0.43	-
pct time queue full	combined better by 9.5pp	-	-	priority better by 7.1pp
pct time under pressure	combined better by 50.0pp	combined better by 17.4pp	combined better by 40.7pp	priority better by 9.3pp
avg length of stay	combined better by 4.01h	combined better by 1.28h	combined better by 2.02h	priority better by 1.99h
rejection rate low	-	-	combined better by 30.7pp	base better by 26.7pp
rejection rate medium	combined better by 15.7pp	-	combined better by 7.8pp	priority better by 7.9pp
rejection rate high	combined better by 26.4pp	combined better by 6.6pp	-	priority better by 22.9pp

7.2 Exponential Arrivals (Demand Stress Test)

Reuses the 30-replication results already generated in Experiment 4, kept as an entirely separate comparison from the uniform-arrival results above.

Table 6: All four scenarios, all 11 tracked metrics, mean and 95% CI across 30 replications each (exponential arrivals).

Scenario–Metric	Mean	Low	High	n
admission rate base	80.90%	79.52%	82.27%	30.0
admission rate 12beds	92.32%	91.46%	93.18%	30.0
admission rate priority	78.18%	76.81%	79.55%	30.0
admission rate combined	90.07%	89.07%	91.07%	30.0
rejection rate base	19.10%	17.73%	20.48%	30.0
rejection rate 12beds	7.68%	6.82%	8.54%	30.0
rejection rate priority	21.82%	20.45%	23.19%	30.0
rejection rate combined	9.93%	8.93%	10.93%	30.0
avg waiting time base	3.77 h	3.59 h	3.96 h	30.0
avg waiting time 12beds	2.17 h	2.01 h	2.32 h	30.0
avg waiting time priority	2.29 h	2.19 h	2.39 h	30.0
avg waiting time combined	1.40 h	1.31 h	1.49 h	30.0
avg bed utilization base	95.07%	94.40%	95.75%	30.0
avg bed utilization 12beds	88.99%	88.00%	89.99%	30.0
avg bed utilization priority	93.97%	93.24%	94.70%	30.0
avg bed utilization combined	87.80%	87.03%	88.57%	30.0
peak queue length base	5.00	5.00	5.00	30.0
peak queue length 12beds	5.00	5.00	5.00	30.0
peak queue length priority	5.00	5.00	5.00	30.0
peak queue length combined	5.00	5.00	5.00	30.0
pct time queue full base	10.03%	9.23%	10.83%	30.0
pct time queue full 12beds	4.64%	4.04%	5.25%	30.0
pct time queue full priority	7.42%	6.69%	8.15%	30.0
pct time queue full combined	3.52%	2.98%	4.07%	30.0
pct time under pressure base	69.49%	66.33%	72.64%	30.0
pct time under pressure 12beds	43.50%	40.59%	46.40%	30.0
pct time under pressure priority	62.72%	60.09%	65.34%	30.0
pct time under pressure combined	39.50%	37.22%	41.78%	30.0
avg length of stay base	27.72 h	27.46 h	27.98 h	30.0
avg length of stay 12beds	26.10 h	25.89 h	26.32 h	30.0
avg length of stay priority	26.75 h	26.55 h	26.95 h	30.0
avg length of stay combined	25.52 h	25.36 h	25.67 h	30.0
rejection rate low base	15.90%	13.75%	18.05%	30.0
rejection rate low 12beds	6.14%	5.03%	7.26%	30.0
rejection rate low priority	36.80%	34.46%	39.14%	30.0
rejection rate low combined	16.99%	15.30%	18.68%	30.0
rejection rate medium base	20.25%	18.89%	21.62%	30.0
rejection rate medium 12beds	8.34%	7.33%	9.36%	30.0
rejection rate medium priority	14.69%	13.43%	15.95%	30.0

Scenario–Metric	Mean	Low	High	n
rejection rate medium combined	6.61%	5.66%	7.56%	30.0
rejection rate high base	30.90%	24.58%	37.22%	30.0
rejection rate high 12beds	12.29%	7.13%	17.45%	30.0
rejection rate high priority	9.54%	5.54%	13.55%	30.0
rejection rate high combined	4.69%	1.46%	7.92%	30.0

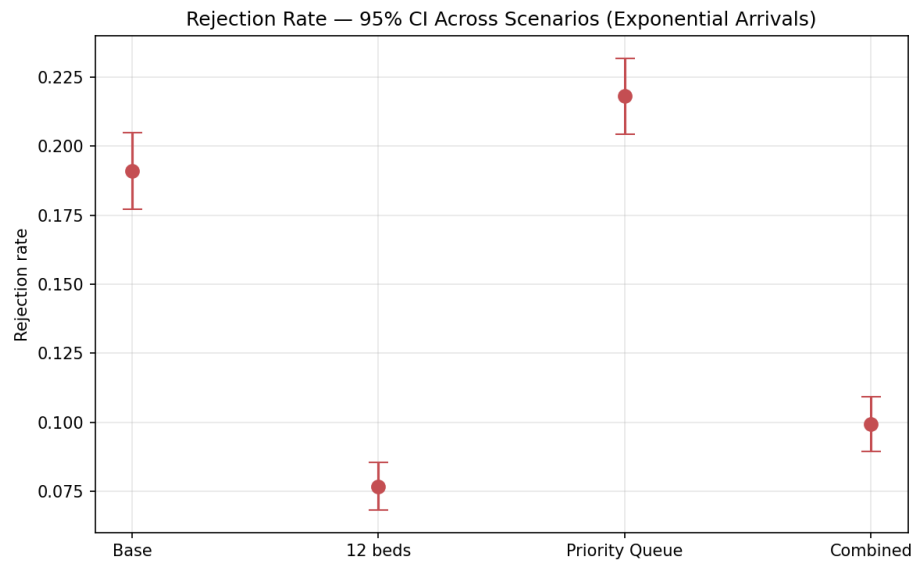


Figure 7: Rejection rate, mean and 95% CI, across all four scenarios (exponential arrivals).

Table 7: Practical significance matrix (exponential arrivals). Same reading rules as the uniform-arrivals significance matrix above: “–” means not practically meaningful; otherwise the better scenario and margin are named. Higher is better for admission rate and bed utilization; lower is better for every other metric.

Metric	Combined vs. Base	Combined vs. Capacity-alone	Combined vs. Queue-Policy-alone	Queue Policy vs. Base
admission rate	combined better by 9.2pp	-	combined better by 11.9pp	-
rejection rate	combined better by 9.2pp	-	combined better by 11.9pp	-
avg waiting time	combined better by 2.38h	-	-	priority better by 1.49h
avg bed utilization	base better by 7.3pp	-	priority better by 6.2pp	-
peak queue length	-	-	-	-
pct time queue full	combined better by 6.5pp	-	-	-
pct time under pressure	combined better by 30.0pp	-	combined better by 23.2pp	priority better by 6.8pp
avg length of stay	combined better by 2.20h	-	combined better by 1.23h	-
rejection rate low	-	12beds better by 10.8pp	combined better by 19.8pp	base better by 20.9pp
rejection rate medium	combined better by 13.6pp	-	combined better by 8.1pp	priority better by 5.6pp
rejection rate high	combined better by 26.2pp	-	-	priority better by 21.4pp

7.3 Interpretation

Under uniform arrivals, the combined scenario is practically better than capacity-alone on several metrics (waiting time, queue pressure, high-priority rejection). Under exponential arrivals, that gap narrows to essentially one metric (`rejection_rate_low`) — once arrivals are bursty, most of the achievable improvement comes from raw capacity, and the priority queue’s marginal contribution on top of it becomes hard to distinguish from noise at $n = 30$ replications. The priority queue *alone* versus base also shows an interesting trade-off in both arrival regimes: `avg_bed_utilization` sometimes favors the base/capacity-alone scenario over the combined one — reordering who is served does not necessarily increase how *busy* the beds are, only who occupies them.

8 Recommendation

1. **Expand capacity to 12 beds.** This is the single intervention with the largest, most consistently practically-significant effect across both arrival regimes, cutting rejection rate from $\sim 15\%$ to $\sim 1.7\%$ under normal demand.
2. **Adopt the priority queue as a fairness safeguard, not a throughput lever.** It does not reduce the overall rejection rate, but it reliably shifts rejection risk away from high-priority patients and onto low-priority ones — valuable on clinical-equity grounds even though it will not show up as an aggregate efficiency gain.
3. **Re-evaluate the priority queue’s marginal value under bursty demand.** Its benefit on top of 12 beds is much smaller when arrivals are exponential rather than uniform; if real arrival data turns out to be closer to exponential, the case for implementing the queue policy should rest primarily on fairness, not on expected throughput gains.
4. **Do not expand beyond 12 beds** without a stronger justification than rejection rate alone — the $12 \rightarrow 14$ bed step already falls below our practical-significance threshold, so further expansion should be justified by other criteria (e.g., financial cost per marginal bed, staffing constraints) not modeled here.

9 Individual Contributions

Team member	Contribution
Bahar Jafarnejad	simulation engine & verification checks & visualizations
Arshia Samimi	experiment design & statistical analysis & report